

University of Wollongong
School of Computing and Information Technology
CSIT121 Object Oriented Design and Programming
Assignment 2

Objectives

- To apply Object Oriented Design (OOD).
- To apply Object Oriented Programming (OOP) using Python.

Note: Please read the submission requirement carefully. Scripts will be used to grade and update your score to Moodle. Failure to name your file name correctly in the required format will result in zero marks updated to Moodle.

Tasks

You are to build a Physiotherapy Session Management System (PSMS) for a clinic. The system will allow patients to book various types of physiotherapy sessions (e.g., general therapy, sports injury, post-surgery rehab, paediatric therapy), and keep records of each session.

Requirements

- All sessions are booked by duration in 30 minutes block.
- Each session type has a base rate per 30 minutes.
- Paediatric therapy is only for age 18 and below.
- Senior (60 years old and above) get a 20% discount on the total cost.
- All costs are rounded to the nearest 10 cents.

Example Rounding:

Calculated Cost	Rounded Cost
\$45.23	\$45.20
\$45.26	\$45.30
\$45.25	\$45.30

Abstract Class: Session

Represents a generic physiotherapy session. This class defines common attributes and an abstract method for cost calculation.

Attribute	Description
__session_id	Unique ID for the session (auto-incremented int, starting from 1)
__patient	A reference to a Patient object.
__duration	Duration of the session in minutes. (in 30 mins block)
__base_rate	Base rate per 30 minutes

Method	Description
calculate_cost()	Abstract method to be implemented by subclasses to compute the session cost based on type-specific rules.

Subclass of Session : GeneralTherapy

Represents a standard physiotherapy session.

Attribute	Description
__focus_area	A string indicating the body part being treated (e.g., "lower back", "shoulder").

Method	Description
calculate_cost()	Calculates cost based on duration and base rate. No discounts or surcharges

Subclass of Session : SportsInjury

Represents a session focused on sports-related injuries.

Attribute	Description
__sport_type	Type of sport associated with the injury (e.g. "soccer", "tennis").
__injury_severity	Severity level (1 to 3)
__pro_athlete	A Boolean to indicate if patient is a national professional athlete.

Method	Description
calculate_cost()	Same as general therapy unless patient is a professional athlete (80% discount).

PostSurgeryRehab (Subclass of Session)

Represents rehabilitation sessions for post-surgery patients.

Attribute	Description
__surgery_type	Type of surgery (e.g., "ACL reconstruction", "hip replacement").

Method	Description
calculate_cost()	Adds a 15% surcharge to the base cost due to specialized care.

Subclass of Session : PaediatricTherapy

Represents therapy sessions for children.

Attribute	Description
<code>__guardian_name</code>	Name of the parent or guardian attending the session

Method	Description
<code>calculate_cost()</code>	Applies a 10% discount to the base cost.

Class: Patient

Stores patient information.

Attributes	Description
<code>__patient_id</code>	Unique ID for the patient (auto-incremented int, starting from 1)
<code>__name</code>	Full name of the patient
<code>__birthdate</code>	Birthdate of the patient -> date type

Method	Description
<code>getAge()</code>	Return the patient's age
<code>__str__()</code>	Returns a formatted string of patient info

Class: ClinicBookingSystem

Manages patients and session bookings.

Attributes: decides on the attributes required. All attributes must be private

Method	Description
<code>add_patient(name, birthdate)</code>	Adds a new patient. Returns the new patient id.
<code>book_session(session_type, patient_id, duration, base_rate, booking_args)</code>	Books a session of the session type. ("General", "Sports", "PostSurgery", "Paediatric") booking_args is a named argument (**kwargs) that is related to that session type: General -> focus_area Sports -> sport_type, injury_severity, pro_athlete PostSurgery -> surgery_type Paediatric -> guardian_name https://www.geeksforgeeks.org/python/args-kwags-python/ Return False if any of the booking data is invalid.
<code>get_patient(patient_id)</code>	Returns the Patient object or False if not found.
<code>get_sessions(patient_id)</code>	Returns a list of sessions for that patient id. Return False if the patent id not found.

Note:

- If an error occur in a method, the method should return a False. Do not throw Exception.
- All instant variables must be private.
- You may implement the variables' getter and setter methods when required.
- You may implement additional methods in each class.

Test Code and automated grading

The testing code StudentAssertionTest.py has been provided to assist you to test your code.

Ensure that you have passed all the test cases before submission, else zero marks will be awarded if the StudentAssertionTest.py failed to run.

Another script with comprehensive test cases will be used to automate the grading of your assignment.

Submission requirements (very important)

- Submit the python code in a txt file containing the code for all the classes. Do not include your test code.
- For full time student:
 - File name must be in the form of: **TXXF_NAME_UOWID.txt** where XX is your tutorial group, NAME is your full name (without space or underscore) and UOWID is your 7-digit UOW ID number (**not SIM student ID number**).
 - For example, **T02F_JeffreyTan_8080426.txt**
- For part time student:
 - File name must be in the form of: **TXXP_NAME_UOWID.txt** where XX is your tutorial group, NAME is your full name (without space or underscore) and UOWID is your 7-digit UOW ID number (**not SIM student ID number**).
 - For example, **T02P_JeffreyTan_8080426.txt**
- **Note**
 - Automated script will be used to extract your file for grading. Failure to follow the above instructions will result in failure to extract your submission correctly.
 - Ensure that the UOW ID number is correct as it will be used by the scripts to update your marks in Moodle. Wrong UOW ID number will result in your marks not updated in Moodle.
 - Note as there are many tutorial groups, failure to name your file name correctly in the above format might result in zero marks.
- Late submission will be penalized 5% per day late.
- Late submission more than 4 days late will be awarded a mark of zero